Project Playbook: Automated Customer Churn Analytics Dashboard Rollout

Created by: Sakshi Rathod | B.Tech Data Science
Methodology: Generative AI Project Management (Persona, Briefing, Skeptic & Process Optimization Patterns)

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1. Project Charter & Scope (Persona Pattern)

Business Objective:
Design and deploy an automated Customer Churn & Retention Analytics Dashboard for an e-commerce platform. The goal is to give the Marketing and Operations teams real-time visibility into at-risk customer segments and reduce quarterly customer churn by 15% without requiring manual SQL queries from the engineering team.

Key Metrics Tracked (KPIs):

- **Monthly Churn Rate (%):** Percentage of customers who stop purchasing over a rolling 30-day window.
- **Customer Lifetime Value (CLV):** Projected revenue from active vs. at-risk cohorts.
- **Retention Cohort Analysis:** Repeat purchase behavior broken down by acquisition channel.
- **Churn Risk Score:** Predictive flagging (High / Medium / Low) based on login frequency and support tickets.

Initial 4-Week Rollout Timeline:

- **Week 1 — Discovery & Data Extraction:** Align with stakeholders on KPI definitions; extract raw historical transaction and user activity logs via SQL.
- **Week 2 — Data Cleaning & Modeling:** Handle missing values, validate data schemas, and establish automated data pipelines.
- **Week 3 — Dashboard UI Development:** Build interactive visual views, filters, and cohort tables for non-technical users.
- **Week 4 — UAT & Stakeholder Sign-Off:** Conduct User Acceptance Testing (UAT) with the Marketing team, refine workflows, and launch live.

Stakeholder Roles & Responsibilities:

- **Technical Product / Project Coordinator (Sakshi Rathod):** Owns timeline execution, risk mitigation, cross-functional communication, and data quality validation.
- **Data Engineering / Analytics Team:** Manages database connections, SQL data transformation, and pipeline stability.
- **Marketing & Retention Lead:** End-user of the dashboard; defines campaign triggers based on churn insights.

2. Dynamic Risk & Schedule Update (Briefing Pattern)

> ***Incoming Project Disruption Trigger — Mid-Project Alert from Data Engineering (Start of Week 2):*** *"30% of the customer transaction logs from the last quarter have missing timestamps and corrupted User IDs. Full data cleaning and recovery will require 4 additional days."*

Updated Project Briefing & Parallel Schedule Adjustment:
To absorb the 4-day data engineering delay without pushing back the final Week 4 launch deadline, the rollout schedule is dynamically updated from a sequential workflow to a parallel execution track:

- **Week 2 (Updated — Parallel Track A):** Data Engineering isolates and repairs the 30% corrupted quarterly logs using cross-referenced payment gateway timestamps.
- **Week 2 (Updated — Parallel Track B):** Instead of waiting for 100% clean data, Dashboard UI development begins immediately using the clean 70% historical dataset as a staging baseline.
- **Week 3 (Updated):** Merge the recovered 30% dataset into the already-built dashboard wireframe; run automated schema validation checks.
- **Week 4 (On Schedule):** Proceed with Marketing UAT and final launch with zero net delay to the business deadline.

Risk Mitigation Matrix:

Identified Risk	Impact Level	Mitigation Strategy
Corrupted User IDs & Missing Timestamps (30% of logs)	High	Implement rule-based data imputation by cross-matching order confirmation emails and payment gateway logs; quarantine unrecoverable rows (<2%) to prevent metric distortion.
Dashboard UI Bottleneck During Data Recovery	Medium	Decouple frontend dashboard wireframing from backend data cleaning by building views on a validated 70% sample dataset.
Skewed Churn Metrics Post-Merge	High	Run automated pre-launch parity checks comparing raw SQL totals against dashboard visual summaries before stakeholder handoff.

3. Skeptic Review & Process Optimization

Critical Assumption Challenge (Skeptic Pattern):
Before final deployment, a skeptical audit of the project assumptions was conducted to prevent post-launch adoption failure:

1. **Assumption 1:** "The Marketing team will check the dashboard every morning and immediately know what action to take."
   - **The Flaw:** Dashboards show *what* is happening, not *what to do*. Busy marketers often suffer from "dashboard fatigue" and stop logging in after two weeks.
   - **The Fix:** Add an automated weekly email digest that pushes the top 100 "High-Risk Churn Users" directly to the Marketing team's inbox every Monday morning as a ready-to-use CSV list.
2. **Assumption 2:** "All stakeholders agree on what 'Churned Customer' means."
   - **The Flaw:** Finance might define churn as 90 days of inactivity, while Marketing defines it as 30 days. Conflicting definitions destroy trust in the data.
   - **The Fix:** Lock in a standardized, documented data dictionary at the top of the dashboard UI (e.g., *Churn = 45 days without a repeat order*) signed off by all department heads.
3. **Assumption 3:** "Manual weekly data refreshes will be easy to maintain."